$$Au + 4H^+ + NO_3^- + 4\,Cl^- \longrightarrow AuCl_4^- + NO + 2H_2O$$
$$3Pt + 16H^+ + 4NO_3^- + 18Cl^- \longrightarrow 3PtCl_3^{2-} + NO + 8H_2O$$

Uses of hydrochloric acid:

1. Hydrochloric acid is used for the manufacture of chlorine, ammonium chloride, glucose from corn starch etc.,
2. Extraction of glue from bone and for purification of bone black

3.3.3 Trends in physical and chemical properties of hydrogen halides:

Preparation:

Direct combination is a useful means of preparing hydrogen chloride. The reaction between hydrogen and fluorine is violent while the reaction between hydrogen and bromine or hydrogen and iodine are reversible and don't produce pure forms.

Displacement reactions:

Concentrated sulphuric acid displaces hydrogen chloride from ionic chlorides. At higher temperatures the hydrogen sulphate formed react with further ionic chloride. Displacement can be used for the preparation of hydrogen fluorides from ionic fluorides. Hydrogen bromide and hydrogen iodide are oxidised by concentrated sulphuric acid and can't be prepared in this method.

Hydrolysis of phosphorus trihalides:

Gaseous hydrogen halides are produced when water is added in drops to phosphorus tri halides except phosphorus trifluoride.

$$PX_3 + 3H_2O \longrightarrow H_3PO_3 + 3HX$$

Hydrogen bromide may be obtained by adding bromine dropwise to a paste of red phosphorous and water while hydrogen iodide is conveniently produced by adding water dropwise to a mixture of red phosphorous and iodine.

$$2P + 3X_2 \longrightarrow 2PX_3$$
$$2PX_3 + 3H_2O \longrightarrow H_3PO_3 + 3HX$$

(where X=Br or I)

Any halogen vapours which escapes with the hydrogen halide is removed by passing the gases through a column of moist red phosphorous.

From covalent hydrides:

The halogens are reduced to hydrogen halides by hydrogen sulphide.

$$H_2S + X_2 \longrightarrow 2HX + S$$

Hydrogen chloride is obtained as a by-product of the reactions between hydrocarbon of halogens.

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